Jiecheng Lu

● NextGen sequence modeling ● Attention meets dynamical systems ● Forecasting for real-world decisions

jlu414@gatech.edu · ljc-fvnr.github.io · Google Scholar · +1 470-929-4016



Ph.D. candidate in Machine Learning at Georgia Tech advised by Shihao Yang. I work on how sequence models read context, store knowledge, and rewrite their own weights, toward self-improving LLMs and agents. My architectures make LLMs and time-series models more expressive under practical compute constraints; I now study parametric memory as a route to continual learning. Since 2024, I have published 10 papers as the main contributor at ICLR, ICML, and NeurIPS.

01 EDUCATION

Ph.D. in Machine Learning, Georgia Institute of Technology, Atlanta, GA (Advisor: Shihao Yang) — Aug 2023–Apr 2027 (expected)

M.S. in Analytics, Georgia Institute of Technology, Atlanta, GA — Aug 2021–Aug 2023

B.S. in Logistics Engineering, Tianjin University, Tianjin, China — Sep 2016–Jun 2020

02 RESEARCH INTERESTS

Parametric memory and continual learning | LLM self-improvement | Agentic AI | Foundation models and LLM pretraining
Efficient and expressive sequence modeling | Latent and in-context reasoning | Time series analysis

03 SELECTED PUBLICATIONS

* corresponding author

● NextGen sequence modeling – LLM pretraining, efficient and expressive attention, and dynamic MLPs

HyperMLP: An Integrated Perspective for Sequence Modeling. **ICML** 2026
Jiecheng Lu, Shihao Yang.

Free Energy Mixer. **ICLR** 2026
Jiecheng Lu, Shihao Yang.

ZeroS: Zero-Sum Linear Attention for Efficient Transformers. **NeurIPS** 2025
Jiecheng Lu, Xu Han, Yan Sun, Viresh Pati, Yubin Kim, Siddhartha Somani, Shihao Yang. ✦ Spotlight

● Attention meets dynamical systems – VAR/ARMA attention and symplectic position encoding beyond RoPE

StretchTime: Adaptive Time Series Forecasting via Symplectic Attention. **ICML** 2026
Yubin Kim, Viresh Pati, Jevon Twitty, Vinh Pham, Shihao Yang, **Jiecheng Lu***.

Linear Transformers as VAR Models: Aligning Autoregressive Attention Mechanisms with Autoregressive Forecasting. **ICML** 2025
Jiecheng Lu, Shihao Yang.

WAVE: Weighted Autoregressive Varying Gate for Time Series Forecasting. **ICML** 2025
Jiecheng Lu, Xu Han, Yan Sun, Shihao Yang.

● Forecasting for real-world decisions – in-context, multivariate, and distribution-valued time series

CAST: Causal Anchored Simplex Transport for Distribution-Valued Time Series. **NeurIPS** 2026
Jiecheng Lu, Jieqi Di, Runhua Wu, Yuwei Zhou.

In-context Time Series Predictor. **ICLR** 2025
Jiecheng Lu, Yan Sun, Shihao Yang.

CATS: Enhancing Multivariate Time Series Forecasting by Constructing Auxiliary Time Series as Exogenous Variables. **ICML** 2024
Jiecheng Lu, Xu Han, Yan Sun, Shihao Yang.

ARM: Refining Multivariate Forecasting with Adaptive Temporal-Contextual Learning. **ICLR** 2024
Jiecheng Lu, Xu Han, Shihao Yang.

04 RESEARCH TRANSLATION

CDC FluSight forecasting hub — Fall 2024–Present

Submit weekly influenza hospitalization forecasts for all U.S. states to the CDC FluSight forecasting hub (entry name: `Gatech-ensemble`), translating time-series forecasting research into public-health decision support. Co-author of Multi-season evaluation and analysis of categorical trend forecasts of influenza hospital admissions in the United States (medRxiv preprint, 2026; submitted to *Nature Communications*).

05 RESEARCH LEADERSHIP AND MENTORING

Since Fall 2024, I have organized a Georgia Tech undergraduate Sequential AI team, leading each project through the full research cycle, from idea to publication.

Yubin Kim	ZeroS (NeurIPS 2025), StretchTime (ICML 2026); 1 ICLR 2027 submission	→ now Ph.D. student in OR, MIT
Viresh Pati	ZeroS (NeurIPS 2025), StretchTime (ICML 2026); 1 ICLR 2027 submission	→ now Ph.D. student in CS, UPenn
Jevon Twitty	StretchTime (ICML 2026); 3 ICLR 2027 submissions	→ undergraduate, graduating 2027
Vinh Pham	StretchTime (ICML 2026); 3 ICLR 2027 submissions	→ undergraduate, graduating 2028

06 TEACHING AND SERVICE

Independent instructor, *ISyE 4031 Regression and Forecasting*, Georgia Tech — Summer 2026

Three sections (one in person, two online), 58 students in 28 project teams. Final Project Showcase; Course Materials. Participant in Georgia Tech's Tech to Teaching program; completed CETL 8717 Course Design for Higher Education.

Reviewing for NeurIPS 2024–2026, ICLR 2025–2026, ICML 2025–2026, and IEEE Internet of Things Journal.

07 SELECTED TALKS

Rethinking Sequence Modeling with HyperMLP: An Integrated Architectural Perspective — Mar 2026

Invited Talk, Knowledge Engineering Group, Tsinghua University (hosted by Prof. Jie Tang, founder of Z.ai)

Rethinking Sequence Modeling: LLM Scaling Laws, Expressivity-Efficiency Tradeoffs, and the Role of Architecture — Feb & Apr 2026

PhD Seminar, Georgia Tech ISyE PhD Student Seminar (Feb) and Georgia Tech Machine Learning Student Seminar (Apr)

08 HONORS AND AWARDS

George Fellowship, Georgia Tech — 2026

Best Poster Award (1st Place), AASF AIX Summit 2026, Poster: "Free Energy Mixer" — 2026

Margaret and Stephen Kendrick PhD Student Fellowship for Research Excellence, Georgia Tech — 2025

M.H. Stewart Fellowship, Georgia Tech — 2023

09 COMPUTING GRANTS

Lambda Research Grant, PI. — Mar 2026–Mar 2027

Subquadratic HyperMLP via Convolution-Based Sequence Mixing.

NVIDIA Academic Grant, awarded project. — Apr 2026–Sep 2026

Scaling Zero-Sum Linear Attention for Cross-Modal Foundation Models.

NSF ACCESS Program, awarded project. — Sep 2025–Sep 2026

Advanced Transformer-based Models for Long-term Time Series Forecasting.

10 PROFESSIONAL EXPERIENCE

Tencent Medical AI Lab (JARVIS Lab) · Data Scientist Intern — Jul 2021–Jul 2022

Led development of deep learning and statistical models for medical insurance forecasting and epidemic monitoring; co-authored 2 medical time-series papers, filed 6 patents as first inventor, and delivered technical solutions for medical facilities across 4 major cities handling up to 10 million data points daily.

Amazon Private Brands, Global Sourcing Team · Business Analyst Intern — Jan 2021–Jul 2021

Applied machine learning and causal inference to cost analysis and sourcing decisions; built AWS-based dashboard web apps that informed decision-making and reported savings of over USD 100,000 monthly.

Peking University, Guanghua School of Management · Research Assistant — Sep 2020–Jan 2021

Country Garden Group · Strategic and Data Analytics Intern — Jul 2020–Sep 2020

11 SELECTED PATENTS

Feature Representation Extraction for Time-Series Data.	CN115114345B
Data Processing for Attention-Based Time-Series Forecasting.	CN114358186B
Training an Anomaly Detection Model for Time-Series Data.	CN115130656B
Interpretable Time-Series Forecasting.	CN114676176B
Multi-Source Multimodal Data Prediction.	CN117010542A
Medical Data Generation from Unordered Inputs.	CN117012347A